

AMENDMENT NO. 2 FEBRUARY 2013
TO
IS 13953 : 1994 *KHANDSARI* — SPECIFICATION

(Page 1, Table 1) — Substitute the following for the existing table:

Table 1 Requirements for *Khandsari*
(Clause 4.5)

Sl No.	Characteristic	Requirement			Method of Test, Ref to	
		<i>Khandsari Desi</i>	<i>Khandsari Sulphur</i>		Annex of this Standard	Clause of IS15279 : 2003
			Grade 1	Grade 2		
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Loss on drying, percent by mass, <i>Max</i>	0.5	0.3	0.5	—	4
ii)	Pol percent, <i>Min</i>	96.0	97.5	96.5	—	5
iii)	Sulphur dioxide (on, dry basis) ppm, <i>Max</i>	Nil	70	70	—	13
iv)	Acid insoluble ash, (on dry basis), percent by mass, <i>Max</i>	0.7	0.3	0.5	A	—
v)	Calcium oxide (CaO), mg/100 g, <i>Max</i>	100	100	100	B	—
vi)	Specific conductivity (mhos/cm ² × 10 ⁶), in 5 percent solution at 30° C, <i>Max</i>	100	—	300	C	—

[Page 2, clause 5.2 (e)] — Substitute ‘Net quantity’ for ‘Net mass’.

[Page 2, clause 5.2 (f)] — Insert the following after Sl No. (f):

‘(g) Any other requirements as specified under the *Legal Metrology (Packaged Commodities) Rules*, 2011 and *Food Safety & Standards (Packaging and Labelling) regulation*, 2011.’

(Page 2, clause A-3.1) — Insert the following ‘Annex B and Annex C’ at the end:

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ANNEX B
[Table 1, Sl No. (v)]
DETERMINATION OF CALCIUM OXIDE (CaO) CONTENT

B-1 APPARATUS

B-1.1 Calibrated Brix Spindle

B-1.2 Brix Cylinder

B-1.3 Conical Flasks, of 250 ml capacity.

B-1.4 Beakers, of 100 or 250 ml capacity.

B-1.5 Funnels

B-1.6 Pipette, calibrated at 10 ml.

B-2 REAGENTS

B-2.1 EDTA solution (Disodium salt of ethylene diamine tetraacetic acid solution) — Weigh accurately 6.647 3 g of EDTA into a beaker; dissolve the same in distilled water and make up the solution to 1 000 ml to obtain exactly M/56 solution.

B-2.2 Ammonia Liquor

B-2.3 Lead Sub Acetate (Horne' Dry Lead)

B-2.4 Potassium Ferrocyanide Powder

B-2.5 Potassium Iodide

B-2.6 Eriochrome Black T — Weigh 0.1 g of Eriochrome Black T into a 100 ml volumetric flask and dissolve the same in rectified spirit or absolute alcohol. Make up the volume with the same solvent and use it as the indicator.

B-3 PROCEDURE

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B-3.1 Take the sugar sample as obtained from the factory and dilute to 15° Bx. Transfer about 150 ml of diluted solution into a conical flask. Clarify the solution by lead subacetate (Adding, if necessary some amount of active carbon). Transfer about 60 ml of the clarified solution into a dry conical flask or flask previously rinsed with the clarified solution. Add powdered potassium ferrocyanide little by little till no further precipitate forms. Shake thoroughly and filter. Test the filtrate with potassium iodide. Collect the lead free filtrate in a conical flask. Pipette out 10 ml of this filtrate into a clean conical flask previously rinsed with distilled water and dried. Add 5 to 6 drops of liquor ammonia and 4 to 5 drops of the indicator solution when a pink red colour appears. Titrate against EDTA solution shaking the contents of the flasks after every addition of EDTA solution. The end point is indicated by the sharp change of colour from red to blue. Note the volume of the titrant.

B-4 CALCULATION

Calcium oxide content, mg/100 g = $V \times 100$ mg/litre of juice or diluted solution.

ANNEX C

[Table 1, Sl No. (vi)]

DETERMINATION OF SPECIFIC CONDUCTIVITY

C-1 FIELD OF APPLICATION

The specific conductivity in solutions gives measure of the concentration of ions at a given temperature, present in solutions. This method is applicable to *khandsari*.

C-2 DEFINITION

C-2.1 Specific Conductivity — *Specific conduction is defined as the conducting power of all the ions present in 1 cm cube of homogeneous solution at a given temperature. As such it gives the conducting power of all the ions present in 1 cm³ of solution.*

C-3 PRINCIPLE

The Specific conductivity of *khandsari* solution at a concentration of 5g/100g is determined at 30°C.

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C-4 REAGENTS

C-4.1 Purified Water — For preparation of all solutions (*Khandsari* and potassium chloride) use twice-distilled or deionized water with a conductivity of less than 1 $\mu\text{S}/\text{cm}$.

C-4.2 Potassium Chloride (0.01 mol/L) — Weigh out 745.5 mg after first dehydrating by heating to 500°C (dull red heat). Dissolve in water in a 1 litre volumetric flask and make up to the mark.

C-4.3 Potassium Chloride (0.002 mol/L) — Dilute 10 ml of potassium chloride solution, 0.01 mol/L (*see C-4.2*) and make it up to the mark in a 500 ml volumetric flask. This solution has a conductivity of $26.6 \pm 0.3 \text{ S}/\text{cm}$ at 30°C (after deduction of the specific conductivity of the water used).

C-5 APPARTUS

C-5.1 Calibrated the Conductivity Meter, with help of solution of potassium chloride (*see C-4.2* and *C-4.3*) at 30°C.

C-5.2 Volumetric Flasks, 100,500 and 1000 ml

C-5.3 Pipettes, 10 ml, conforming to class A of IS 1117.

C-5.4 Analytical Balance, capable of weighing to the nearest 1.0 mg.

C-6 PROCEDURE

C-6.1 Dissolve 5.0 g of sugar in water (*see C-4.1*) in a 100 ml volumetric flask and make up to volume at 30°C.

C-6.2 After thorough mixing, transfer the solution into the measuring cell and measure the specific conductivity at 30°C. Check the measurement using the reference solution (*see C-4.3*).

C-7 EXPRESSION OF RESULT

Report the value of specific conductivity of *khandsari* at 30°C in the units of $\mu\text{S}/\text{cm}$

(FAD 2)

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